C++ Notes: While Loops

(while vs do-while, infinite loop warning)

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Learning Objectives

After this section, you will be able to:

- Write a `while` loop with a condition
- Understand loop execution flow (check condition each iteration)
- Avoid infinite loops
- Understand `do-while` (executes once before checking condition)

Core Idea

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A loop repeats a block of code.
A `while` loop repeats **while the condition is true**. Each iteration:

1. check condition
2. run the block (if condition is true)
3. go back and check again

1. Basic while loop

Listing 1: Print 1..5 using while

```
int index = 1;

while (index <= 5) {
  cout << index << endl;
  index++;
}
```

Output:

```
1
2
3
4
5
```

Execution flow (easy to remember)

1. Start with `index = 1`
2. Check `index <= 5`
3. Print `index`
4. Increment `index`
5. Repeat until condition becomes false

2. Infinite loop warning

Infinite loop

If the condition never becomes false, the loop runs forever.
Example (dangerous): if you forget `index++`, `index` stays 1 forever.

Listing 2: Example infinite loop (do not run)

```
int index = 1;

while (index <= 5) {
  cout << index << endl;
  // index++; // missing => infinite loop
}
```

3. When while loop runs zero times

Listing 3: If condition is false at the start, while runs 0 times

```
int index = 6;

while (index <= 5) {
  cout << index << endl;
  index++;
}
```

Output: (nothing printed)

4. do-while loop

do-while

A `do-while` loop runs the body first, then checks the condition. So it always runs at least once.

Listing 4: do-while runs once even if index starts at 6

```
int index = 6;
```

```
do {
  cout << index << endl;
  index++;
} while (index <= 5);
```

Output:

6

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